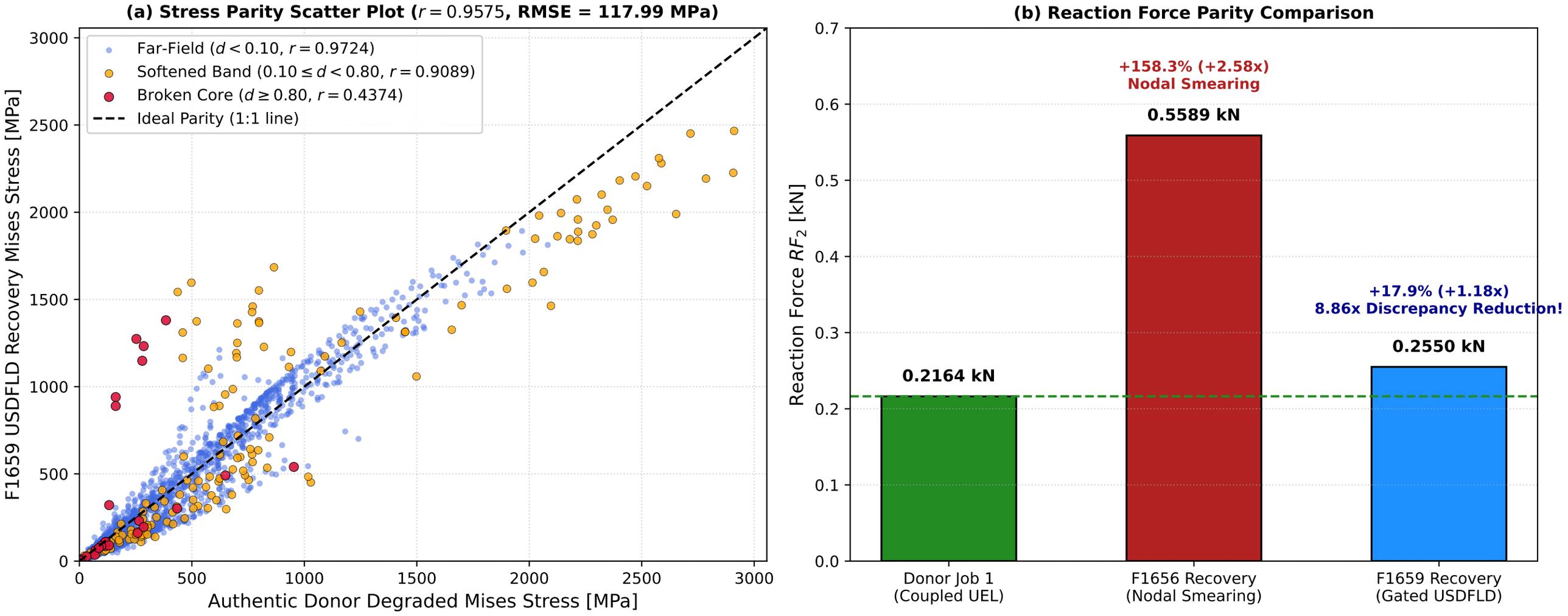

(a) Stress Parity Scatter Plot ($r = 0.9575$, RMSE = 117.99 MPa)
Far-Field ($d < 0.10$, $r = 0.9724$)
Softened Band ($0.10 \leq d < 0.80$, $r = 0.9089$)
Broken Core ($d \geq 0.80$, $r = 0.4374$)
Ideal Parity (1:1 line)
F1659 USDFLD Recovery Mises Stress [MPa]
Authentic Donor Degraded Mises Stress [MPa]
0
500
1000
1500
2000
2500
3000
(b) Reaction Force Parity Comparison
Reaction Force RF_2 [kN]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
+158.3% (+2.58x)
Nodal Smearing
0.5589 kN
+17.9% (+1.18x)
8.86x Discrepancy Reduction!
0.2550 kN
0.2164 kN
Donor Job 1
(Coupled UEL)
F1656 Recovery
(Nodal Smearing)
F1659 Recovery
(Gated USDFLD)